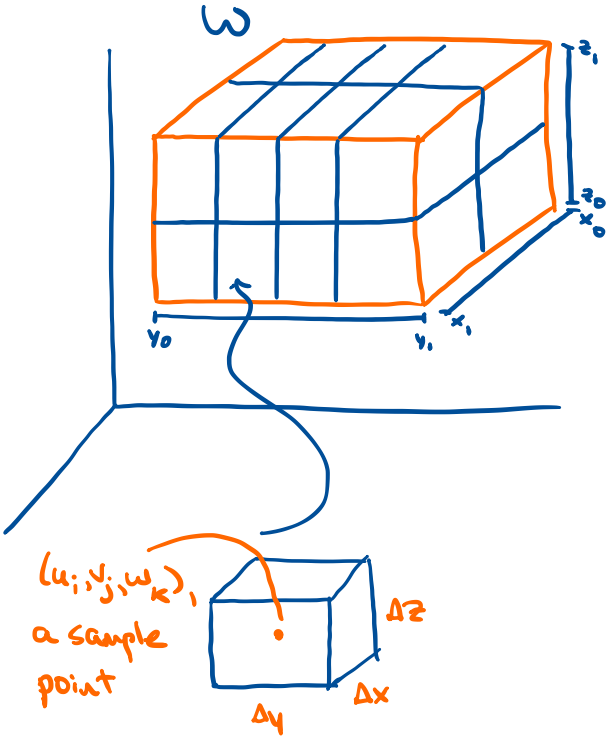

Suppose W is a box in space and $w = f(x,y,z)$ is a function. Divide W into smaller sub-boxes of dimensions Δx, Δy, by Δz. Then the signed "hypervolume" of the "hypercuboid" between the graph of f and the $(i,j,k)^{th}$ sub-box is

$$f(u_i, v_j, w_k)\,\Delta V$$

where $\Delta V = \Delta x\,\Delta y\,\Delta z$ and (u_i, v_j, w_k) is a point in the sub-box. Hence the total signed "hypervolume" between the box W and the graph of f is

$$\sum_{i,j,k} f(u_i, v_j, w_k)\,\Delta V. \quad \leftarrow \text{A Riemann sum}$$

This becomes exact in the limit:

$$\int_W f\,dV = \lim_{\Delta V \to 0} \sum_{i,j,k} f(u_i, v_j, w_k)\,\Delta V.$$

Definition of integral.

Fact: If $f(x,y,z)=1$ for all (x,y,z), then

$$\int_W f\,\partial V = \underbrace{\int_W \partial V = \text{volume of } W}$$

Summary

- $y=f(x)$ is integrated against ∂x ← line element
- $z=f(x,y)$ is integrated against ∂A ← area element
- $w=f(x,y,z)$ is integrated against ∂V ← volume element

Triple integrals can occur over more than just boxes. For example we can have

$$\int_{x_0}^{x_1} \int_{y=g_0(x)}^{y=g_1(x)} \int_{z=h_0(x,y)}^{z=h_1(x,y)} f(x,y,z)\,\partial z\,\partial y\,\partial x.$$

The functions in the limits of the integrals must be of the variables toward the outside. Another example:

$$\int_{y_0}^{y_1} \int_{z=g_0(y)}^{z=g_1(y)} \int_{x=h_0(y,z)}^{x=h_1(y,z)} f(x,y,z)\,\partial x\,\partial z\,\partial y.$$

→ Do all problems.